Relative Adrenal Insufficiency in the Preterm Infant

Cristina E. Gutierrez, MD,* Theodore De Beritto, MD, MS*

*Division of Neonatology, Department of Pediatrics, Mattel Children's Hospital, David Geffen School of Medicine at UCLA, Los Angeles, CA

EDUCATION GAPS

1. Relative adrenal insufficiency has been investigated and identified in the preterm infant with sepsis.
2. More research is necessary to determine if corticosteroids should be used as first-line treatment in infants with suspected relative adrenal insufficiency.
3. More long-term studies are needed to characterize the normal values of cortisol in premature infants.

OBJECTIVES *After completing this article, readers should be able to:*

- Define relative adrenal insufficiency
- Recognize the different zones of the adrenals and their function
- Identify key hormones produced by the adrenal glands
- Evaluate specific barriers to identifying and treating adrenal insufficiency in the preterm infant

ABSTRACT

Identifying relative adrenal insufficiency in the critically ill preterm neonate is not always clear-cut. Preterm infants with vasopressor-resistant shock may have persistent cardiovascular insufficiency, which can result in rapid decompensation. After attempts of resuscitation with fluids and inotropes, these infants are often found to respond to glucocorticoids. This raises the important question of how prevalent adrenal insufficiency is in the preterm population. This article reviews the development and role of the adrenal glands, defines relative adrenal insufficiency in the preterm population, discusses barriers to determining this diagnosis, and describes treatment options.

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ABBREVIATIONS

ACTH	adrenocorticotropic hormone
CRH	corticotropin-releasing hormone
DHEA	dehydroepiandrosterone
HPA	hypothalamic-pituitary-adrenal
PDA	patent ductus arteriosus
3β-HSD	3-β dehydrogenase
11β-HSD1	11-β hydroxysteroid dehydrogenase type 1
11β-HSD2	11-β hydroxysteroid dehydrogenase type 2

INTRODUCTION

When treating infants in the NICU, clinicians are often faced with complex issues and multifactorial problems. This is especially true when diagnosing sepsis in the preterm or low-birthweight infant. When preterm infants develop cardiovascular insufficiency that does not resolve with fluid and inotropic support, it is important to consider relative adrenal insufficiency as a primary cause so that appropriate treatment can be provided. Untreated adrenal insufficiency can be detrimental to the response infants mount to any stressor, particularly sepsis.

In this review, we will describe the embryologic and postnatal development of the adrenal glands. In addition, we will discuss the hypothalamic-pituitary-adrenal (HPA) axis and the hormones produced within this system, as well as the critical roles that they play. We will explore the importance of recognizing and diagnosing relative adrenal insufficiency in the preterm infant, and discuss its impact on the preterm population, specifically in neonatal sepsis. Lastly, we will summarize the most current diagnostic practices for determining relative adrenal insufficiency in the preterm neonate, as well as treatment options.

SEPSIS IN THE PRETERM INFANT

Preterm infants with either early-onset sepsis (occurring before 48–72 hours of age) or late-onset sepsis (occurring after 48–72 hours of age) typically present with poor feeding, temperature instability, respiratory distress, and lethargy. Although antibiotics are an important part of the treatment for sepsis, affected infants must also be monitored for cardiovascular insufficiency, which should be treated promptly and effectively.

In full-term neonates and children, the 2020 sepsis guidelines established in the Surviving Sepsis Campaign recommend fluid boluses in cases of shock and further treatment with inotropes if blood pressure control remains refractory to fluids. (1) However, currently there are no guidelines for management of sepsis in the premature infant, as no large-scale trials have been conducted to address the most effective management in this population. In clinical practice, preterm infants with septic shock are typically treated similarly, first with fluid resuscitation, and if the shock is still present, then using dopamine, norepinephrine, or epinephrine. (2) However, fluid resuscitation in this population must be used judiciously because excessive fluid administration can exacerbate chronic lung disease, delay closure of the patent ductus arteriosus (PDA), increase the risk of intraventricular hemorrhage, and most importantly, increase mortality risk. (3)

Preterm infants with septic shock also differ from their term counterparts in their stress response. The adrenal glands of term infants with stress-induced sepsis release an appropriate amount of cortisol; however, some preterm infants cannot mount an adequate response corresponding to the severity of the illness, known as "relative adrenal insufficiency." (4) If this diagnosis is not recognized in affected preterm infants with shock, these infants may have systemic consequences and potential death.

HYPOTENSION AND SHOCK

The normal blood pressure in preterm infants is highly variable because of the heterogeneity of the population, and there is no well-defined normal range for different postmenstrual ages. The most widely accepted definition for a normal mean blood pressure in this population is 2 to 3 mm Hg above the infant's postmenstrual age in weeks. (3) Although the terms "hypotension" and "shock" are often used interchangeably, these diagnoses are distinct. Hypotension is defined as a low blood pressure compared to a normal value based on gestational and chronologic age. In contrast, shock is defined as inadequate organ tissue perfusion. (2)

Many pathophysiologic mechanisms of hypotension are found in preterm infants with sepsis. Affected infants have impaired cardiac contractility as well as reduced intravascular volume. Both term and preterm infants can experience vasodilation, capillary leak, and fluid losses that lead to shock. (2) However, because the preterm heart has immature myocardium, the preterm infant is less responsive to both endogenous and exogenous catecholamines, making it difficult to treat hypotension. (3) In addition, many preterm infants have a PDA, which can further contribute to decreased cardiac output as a result of diastolic run-off. This constellation can lead to an inability to deliver sufficient oxygen to the tissues, resulting in shock.

Although cardiovascular insufficiency is a large contributor to neonatal decline during periods of stress, it is important to consider other contributing factors that could prevent an appropriate cortisol response, such as immature development of the adrenal glands. One study demonstrated that premature infants receiving inotropic therapy had lower cortisol levels than those who did not, indicating a breakdown along the HPA axis. (4) This could be due to myriad reasons, including cytokine suppression of cortisol synthesis or adrenocorticotropic hormone (ACTH) release, limited adrenal cortical response to the ACTH produced, or insufficient perfusion of the adrenal glands. (5) This lack of circulating cortisol can provide some insight into management of shock secondary to sepsis.

DEVELOPMENT OF THE ADRENAL GLANDS

The initial steps of adrenal cortex development occur during week 4 of gestation when the coelomic epithelium thickens. (6) The cells from the epithelium then invade the mesenchyme and the most medial of these cells form the adrenal primordium at approximately 33 days after conception. The gland then differentiates into the adrenal cortex as well as the fetal zone (inner-most zone) and the definitive zone. This occurs at 50 to 52 days after conception. At this time, the fetal zone is producing high levels of steroidogenic enzymes. At 9 weeks of gestation, the adrenal gland is vascularized and the mesenchymal capsule forms. (6)

The adrenal medulla develops concurrently with the cortex but is derived from neural crest cells. (6) In forming the medulla, cells invade ventrally into the anterior sclerotome and onto the dorsal aorta. At this point, it is now of sympathoadrenal lineage and can start producing catecholaminergic neuronal progenitor cells. Some of these cells travel from the dorsal aorta to the adrenal primordium and become chromaffin cells that are scattered throughout the adrenal cortex. (6)

In addition to the medulla, the main components of the mature adrenal gland are the zona reticularis, zona fasciculata, and zona glomerulosa. However, in the early stages of development, these sections are known as the fetal zone, transitional zone, and definitive zone, respectively. (6) When a fetus reaches term development, the fetal zone contributes the most to the size of the adrenal gland. The fetal zone produces most of the steroidogenic enzymes during this period of early development. This zone involutes postnatally, resulting in a 50% decrease in the size of the adrenals by 2 weeks of age as well as a decrease in androgen production. Its successor, the zona reticularis, begins to form around 3 to 6 years of age, at which point adrenal androgen production restarts. The transitional zone produces cortisol and slowly develops into the zona fasciculata at 3 years of age, whereas the definitive zone becomes the zona glomerulosa. (6) In early development, the zona fasciculata synthesizes cortisol in response to ACTH from the pituitary (in response to corticotropin-releasing hormone [CRH] from the hypothalamus). (4)

MATERNAL-FETAL HPA AXIS

Three main components need to be considered in fetal development of the HPA axis: the pregnant woman, the placenta, and the fetus. (7) Cortisol is initially able to pass through the placenta from the pregnant woman, which suppresses fetal cortisol production through negative feedback of the fetal HPA axis. The placenta acts as a gatekeeper for the maternal cortisol to access the placenta via the enzyme 11-β hydroxysteroid dehydrogenase type 2 (11β-HSD2). This enzyme oxidizes maternal cortisol to inactive cortisone; however, early on in fetal development, the activity of this enzyme is very low and maternal cortisol is able to exert negative feedback on the fetal HPA axis. (7) Thus, the fetal HPA axis is suppressed during early gestation and maternal cortisol is the primary source of steroid the fetus is exposed to (Fig 1).

In the placenta, there is marked production of CRH, with the highest production occurring close to term; this

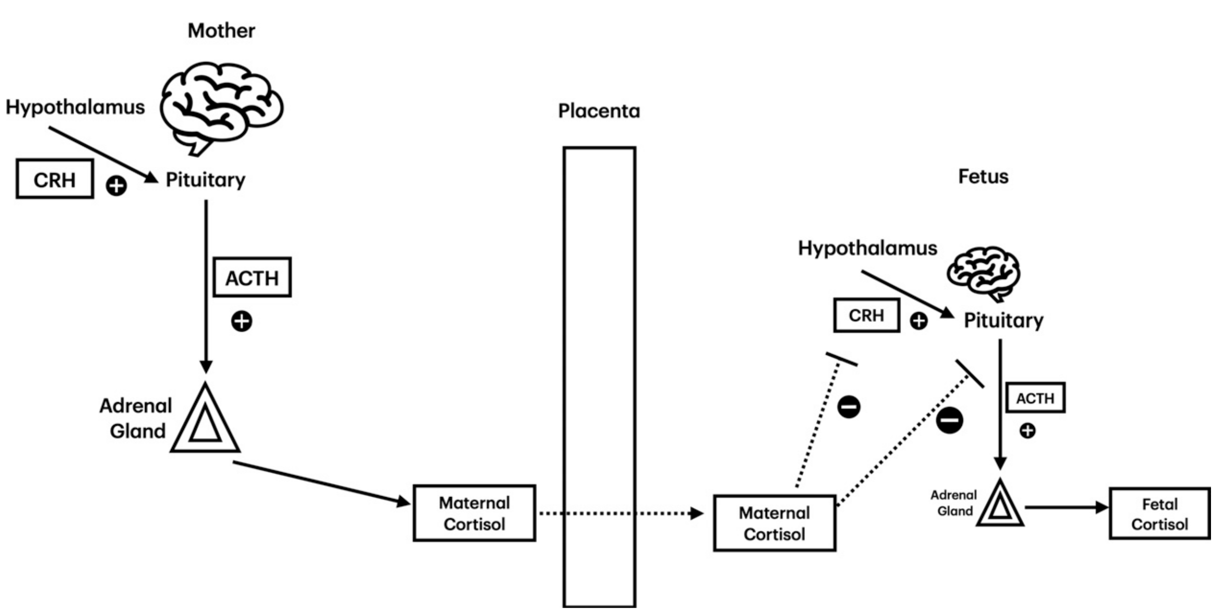

Figure 1 In early gestation, maternal cortisol passing through the placenta inhibits corticotropin-releasing hormone (CRH) production by the fetal hypothalamus as well as adrenocorticotropic hormone (ACTH) production by the fetal pituitary gland. In this way, fetal cortisol production is inhibited until closer to term development.

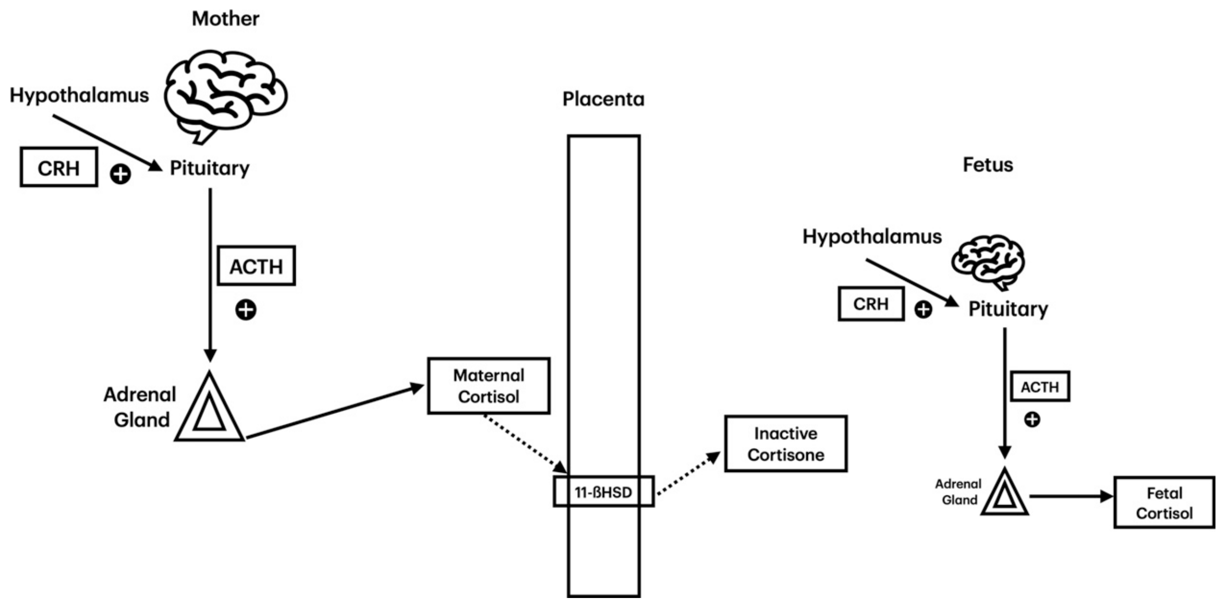

Figure 2 Conversely to early gestation, in later gestation, there is greater presence of 11-β hydroxysteroid dehydrogenase (11β-HSD) in the placenta. Active maternal cortisol is transformed into inactive cortisone by this enzyme. As a result, the fetal hypothalamic-pituitary-adrenal axis is now able to produce fetal cortisol without inhibition.

placental CRH is found in both the maternal and fetal circulations. (7) Before 30 weeks of gestation, the progesterone from the placenta bypasses 3-β dehydrogenase (3β-HSD) to produce cortisol. (4) There is limited 3β-HSD in early gestation and cortisol is not synthesized de novo by the fetus until 30 weeks of gestation. (4)

The activity of 11β-HSD2 ramps up in the third trimester, decreasing the amount of maternal cortisol present in its active form, which spurs the fetus to begin to produce its own fetal ACTH and cortisol. (7) At late gestation, maternal cortisol is no longer predominant in the fetus; when this occurs, ACTH synthesis and fetal cortisol production are increased (Fig 2). (7)

Premature infants are missing out on these key developmental steps of cortisol production, which subsequently contributes to their poor hemodynamic regulation in times of stress and their inability to mount an appropriate stress response. By closely examining the embryologic development of the adrenal glands, differences between preterm and term infants as they develop their respective responses to stress can be better understood.

HORMONES PRODUCED BY THE ADRENAL GLANDS AND THEIR ROLES

Zona Fasciculata

Cortisol is formed in the adrenal cortex from cholesterol stored in the zona fasciculata in response to ACTH. The cortisol produced can bind to either albumin or corticosteroid-binding globulin. Inactive cortisol can be activated via 11-β hydroxysteroid dehydrogenase type 1 (11β-HSD1) and reconverted by 11β-HSD2 in both the pancreas and kidney. (8) Cortisol is a multifunctional hormone regulating metabolism, the stress response, immune function, and the inflammatory response. (8)

In situations of stress, the amygdala processes the emotional stimuli and sends a message to the hypothalamus to activate the sympathetic nervous system. This induces the adrenal glands to release catecholamines. (8) Initially, epinephrine is released and if the stress response persists, the hypothalamus prompts cortisol to be released from the adrenal cortex via the HPA axis. (8)

Cortisol increases blood glucose availability to the brain via gluconeogenesis in the liver, pancreas, adipose tissue, and muscle. (8) It also increases activity of glucagon, catecholamines, and epinephrine. Under nonstressful conditions, cortisol remains protein bound to albumin or cortisol-binding globulin. When cortisol is released and binds to glucocorticoid receptors, it can have a direct impact on the efficacy of catecholamines as it inhibits the breakdown of β receptors and promotes their density. (4) In this way, cortisol promotes hemodynamic stability.

Two primary clinical consequences of insufficient cortisol production in the preterm infant are chronic lung disease and cardiovascular instability. Cortisol helps to suppress inflammation in the developing airway; without this, inflammation can be uncontrolled and the likelihood of developing bronchopulmonary dysplasia increased. Cardiovascular instability related to lack of cortisol can lead to hypotension that is poorly responsive to vasopressors and has the potential for eventual cardiovascular collapse. (9)

Zona Glomerulosa

Aldosterone is a mineralocorticoid made in the zona glomerulosa. When released, the kidneys absorb sodium and excrete potassium. Aldosterone helps to control blood pressure by increasing or decreasing the extracellular fluid volume via sodium regulation. (10) This hormone also helps to regulate the pH in the body, as hydrogen ions are eliminated via exchange with potassium in the nephron's lumen. (10)

Though well-adapted in term infants, the exchange of potassium ions for hydrogen ions can result in alkalosis, which occurs more often in the preterm infant because of immature kidneys. (10) Hyponatremia is another sequela related to the decreased responsiveness to aldosterone at the distal renal tubule and has the potential to contribute to circulatory compromise via low intravascular volume. (9)

Zona Reticularis

Dehydroepiandrosterone (DHEA) and androgenic steroids are produced by the zona reticularis. These hormones serve as the precursors to estrogens and androgens and are weak male hormones. DHEA also contributes to reduced inflammation, improved blood flow, cellular immunity, insulin sensitivity, and bone metabolism. (11) DHEA is principally produced in the fetal zone before it involutes. (12) DHEA is sulfated in the adrenal glands to form DHEA-S, which promotes additional 11β-HSD2 production from the placenta. (7) Higher levels of 11β-HSD2 encourage fetal cortisol production. (7)

Adrenal Medulla

Epinephrine and norepinephrine are produced in the adrenal medulla and are released by chromaffin cells. Norepinephrine, the precursor to epinephrine, causes increased contractility of the heart, increased heart rate, and improved vascular tone. In small doses, epinephrine increases renin release, improves cardiovascular contractility, increases heart rate, and has an effect on bronchodilation. (13) In large doses, epinephrine can cause pupillary dilation, vascular smooth muscle contraction, and contraction of the intestinal sphincter muscle. (13)

Catecholamines are released in cases of acute stress and likely to be elevated during sepsis in term infants. The adrenal medulla is not fully developed in preterm infants, which decreases the likelihood of an appropriate stress response and can lead to cardiovascular compromise, issues with thermoregulation, and poor adaptation in extrauterine life. (14)

DIAGNOSIS

The serum cortisol level is a key marker that is higher during periods of stress and severe illness. (4) Although studies have determined normal cortisol levels in adult patients, the specific normal values in preterm infants have been debated intensely. However, we do know that preterm infants have a lower level of circulating cortisol at baseline. As a result, when preterm infants develop sepsis, hypotension is more likely because of inadequate cortisol levels. In this scenario, administration of glucocorticoids will often improve the infant's blood pressures. (15) Previous ultrasound studies looking at adrenal size as a predictor for adrenal insufficiency have noted that immature adrenals that do not undergo involutional change 3 weeks after birth are at risk for adrenal insufficiency. (9)

Given the overall lack of consensus about diagnostic criteria and the multifactorial picture of stress in this population, relative adrenal insufficiency is difficult to diagnose in the preterm infant. However, in a preterm infant with sepsis who has vasopressor-resistant hypotension or recent surgery, relative adrenal insufficiency should be on the differential. (16) Relative adrenal insufficiency should also be considered in infants born to mothers who recently received antenatal steroids, as this can suppress postnatal adrenal function in the neonate. (9) Affected infants can present with oliguria, hyperkalemia, and hyponatremia that does not typically improve with volume expansion. (9) Obtaining cortisol levels is not typically helpful in this scenario as the results are not available quickly and normal values are not known, so treatment is often pursued based on suspicion alone.

TREATMENT AND OUTCOMES

The initial step in the treatment of shock is fluid resuscitation and subsequent initiation of vasopressors. If the signs of shock do not resolve, it is appropriate to consider relative adrenal insufficiency and administer glucocorticoid therapy. It is important to note that at birth, preterm infants have a limited number of α receptors, an immature myocardium, fewer β receptors, and less adrenocortical reserve compared to older patients. (2) Glucocorticoids help to promote expression of cardiac adrenergic receptors, thereby improving cardiac output. (9) One study examined very-low-birthweight preterm neonates who had refractory hypotension even after they were treated initially with fluid and dopamine. (2) In this study, 1 group was also given hydrocortisone, compared to a control group that only received fluid and dopamine. The group given hydrocortisone had a quicker recovery of blood pressures; however, there was no difference in clinical outcome between the 2 groups. (2) An important factor to acknowledge is that administration of glucocorticoids, such as dexamethasone and hydrocortisone, can suppress the neonate's HPA function; however, recovery can be seen 1 to 2 months after administration, so these medications should not be avoided if they are needed. (12)

There is much debate on the type of steroid that should be used to treat infants with relative adrenal insufficiency. In the 2016 PREMILOC trial comparing dexamethasone, hydrocortisone, and placebo to prevent bronchopulmonary dysplasia, the use of hydrocortisone was favored over dexamethasone, because hydrocortisone has less of an effect on growth and neurodevelopment and influences both mineralocorticoids and glucocorticoids. (17) Although hydrocortisone may improve hemodynamic instability, it can be associated with gastrointestinal perforation in preterm infants receiving indomethacin for PDA closure, so it should be avoided in this group. (18)

The use of steroids as first-line medication in preterm neonates with signs of relative adrenal insufficiency has also been studied. The 5 studies conducted in preterm neonates averaging between 23 and 29 weeks' gestational age were testing whether prophylactically treating adrenal insufficiency would improve associated outcomes such as bronchopulmonary dysplasia. (19) These studies ultimately showed higher serum cortisol results in infants receiving hydrocortisone prophylaxis compared to the control group, which did not receive glucocorticoids, as well as decreased odds of bronchopulmonary dysplasia. (19)

CONCLUSION

In the preterm infant, it is important to monitor all clinical findings of circulatory sufficiency, including blood pressure, heart rate, and urine output. It is difficult to assess the cause of cardiovascular compromise in this population, as we are limited by the lack of standardized blood pressure guidelines and normal cortisol level ranges. Furthermore, the use of pain control in the preterm population can dampen the infant's response to stress and blood pressure regulation by decreasing the amount of stress hormone present. (12) Many factors can contribute to adrenal insufficiency in the preterm infant, including immature adrenal glands, decreased response to CRH, and an impaired ability to produce cortisol. Infants who exhibit shock that does not respond to fluid resuscitation or vasopressors should be considered candidates for hydrocortisone; however, care should be taken to avoid this medication in high-risk populations—particularly infants being treated for a PDA.

Given the poor response to fluids and inotropes in the preterm infant with adrenal insufficiency, there is an opportunity to further study whether corticosteroid administration would be the most efficacious first-line treatment for preterm neonates with shock. By understanding the limitations of the preterm adrenal glands, as well as the clinical signs and symptoms of relative adrenal insufficiency, we can aim to provide infants who have relative adrenal insufficiency with the rapid intervention and support that they need, in an effort to produce the best outcomes.

American Board of Pediatrics Neonatal-Perinatal Content Specification

- Know the pathophysiology of a term or preterm infant with a condition affecting the systemic blood pressure, such as hypotension or hypertension.

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